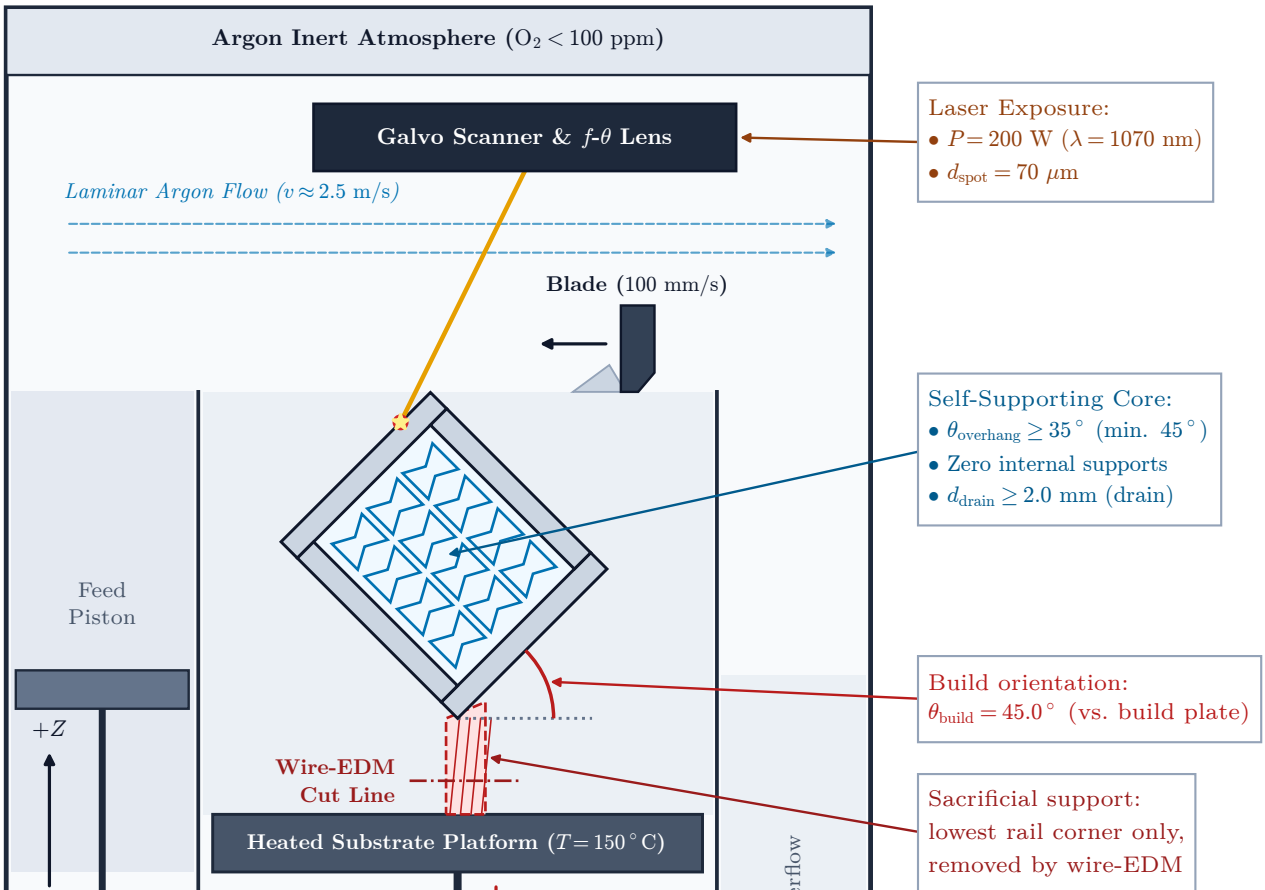
(a) L-PBF Build Chamber & 45° Orientation
Argon Inert Atmosphere ($O_2 < 100$ ppm)
Galvo Scanner & f-θ Lens
Laminar Argon Flow ($v \approx 2.5$ m/s)
Blade (100 mm/s)
Feed Piston
$+Z$
Wire-EDM Cut Line
Heated Substrate Platform ($T = 150\,^\circ$C)
$-Z$ ($\Delta z = 30$ μm)
Overflow
Laser Exposure:
• $P = 200$ W ($\lambda = 1070$ nm)
• $d_{spot} = 70$ μm
Self-Supporting Core:
• $\theta_{overhang} \geq 35\,^\circ$ (min. $45\,^\circ$)
• Zero internal supports
• $d_{drain} \geq 2.0$ mm (drain)
Build orientation:
$\theta_{build} = 45.0\,^\circ$ (vs. build plate)
Sacrificial support:
lowest rail corner only,
removed by wire-EDM